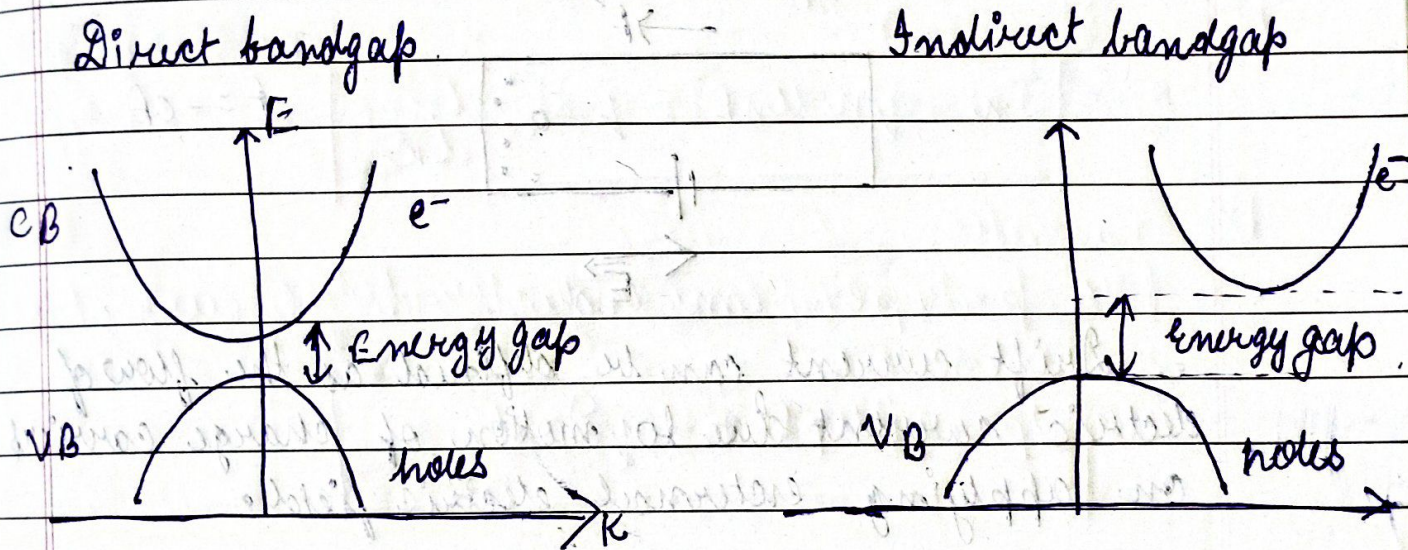


Semiconductor physics

Direct and indirect band-gap Semiconductors:-

The energy of an e^- is given by $E = \frac{p^2}{2m} = \frac{\hbar^2 k^2}{2m}$ where p is momentum, m is mass of an e^- , $\hbar$ is planck's const. and k is propagation constant.

Thus $E \propto k^2$ which is an eqⁿ of parabola. The graph of E vs k :-



i) Maxima of VB & minima of CB occurs at same temp.

Maxima of VB & minima of CB occurs at different values of momentum.

ii) Momentum remains conserved & photon (light) is emitted.

Momentum is not conserved so to conserve it, phonon (heat) is emitted.

iii) They are known as direct band gap semiconductor.

They are known as indirect band gap semiconductors.

iv) Used in LED & Laser diodes

used in amplifier, rectifier & transistors.

v) Ex:- Compound Semiconductors
Ga As C Gallium Arsenide

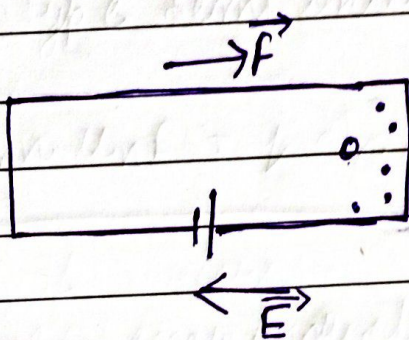
ex:- Intrinsic S.C
Si, Ge.

1) Drift current :-

$$\text{Drift velocity } (V_d) = \mu E$$

It is the velocity attained in a certain direction by charge carriers on applying $\vec{E}$ across the semiconductor material.

This combined movement of charge carriers (e^- & h^+) constitutes a current called as Drift current.



$$F = -eE$$

$\therefore$ Drift current can be defined as the flow of electric current due to motion of charge carriers on applying external electric field.

$$I = neAV_d$$

$$\frac{I}{A} = neV_d$$

$$J = ne\mu E$$

J (current density for e^- & holes)

$$\begin{aligned} J_n &= nq_n \mu_n E \text{ A/cm}^2 \\ J_p &= p q_p \mu_p E \text{ A/cm}^2 \end{aligned}$$

(Here n & p are e^- & hole densities)

$$\therefore \text{Combined } J = nq_n \mu_n E + p q_p \mu_p E$$

$$J = (n\mu_n + p\mu_p)qE$$

$$\therefore \begin{cases} J = \sigma E \\ J = \underbrace{(n\mu_n + p\mu_p)qE}_{\sigma} \quad A/cm^2 \end{cases}$$

2) Diffusion current:-

If no. of e^- s or holes is greater in one region of a semiconductor in comparison to rest of the region, then we say that concentration gradient exists in the material.

So the movement of charge carriers of same type takes place from a region of higher conc. to lower conc. current resulting due to this movement of charge is Diffusion current.

Diffusion current density due to e^- s:-

$$J_n = q, I_n \frac{dn}{dx} \quad A/cm^2$$

Diffusion current density due to holes:-

$$J_p = -q, I_p \frac{dp}{dx} \quad A/cm^2$$

Here $\frac{dn}{dx}$ & $\frac{dp}{dx}$ are concentration gradients for e^- & holes resp.

D_n & D_p are diffusion coeff. for e^- s & holes

Total current (Drift + Diffusion)

Total current in a semiconductor
= Drift current + Diffusion current.

$\therefore$ For N-type semiconductor total current is:-

$$J_n = q n \mu_n E + q D_n \frac{dn}{dx}$$

For p-type semiconductor total current is:-

$$J_p = q p \mu_p E - q D_p \frac{dp}{dx}$$

Mobility (μ): -

Electric force on an e^- $F = qE$

Newton's second law $F = ma$

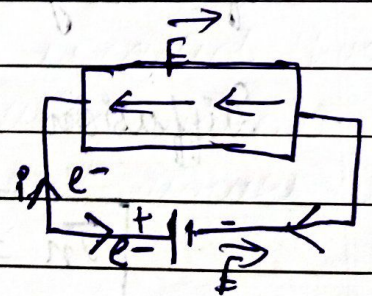
$$a = \frac{F}{m}$$

$$a = \frac{qE}{m} \text{ for one } e^-$$

$\rightarrow$ For all e^- s

$$\int a \, dt = \int \frac{qE}{m} \, dt$$

$$\int \frac{dv}{dt} \, dt = \frac{qE}{m} \int dt$$



$$m \frac{dv}{dt} = qE \quad \text{--- (1)}$$

$$V = \frac{qE}{m} t + C \quad \text{--- (2)}$$

$$t=0, V=0$$

$$0 = \frac{qE}{m} (0) + C$$

$$C = 0$$

Putting value of C in eqⁿ (2)

$$\therefore V = \frac{qE}{m} t$$

$$V = \left(\frac{q \cdot t}{m} \right) E$$

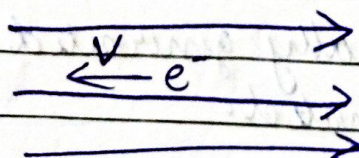
$$\therefore V \propto E$$

$$\therefore V = \mu E \quad \text{Ans}$$

$$\mu = \frac{V}{E}$$

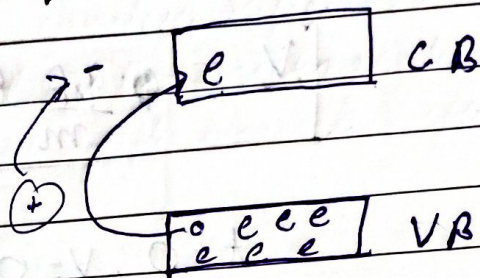
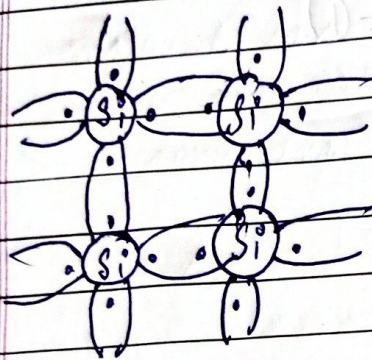
$$\text{SI unit } \frac{\text{m}^2}{\text{Vt-sec}}$$

Mobility of charge carrier is defined as its average velocity under the influence of a unit electric field strength.



$$E = 500 \text{ N/C}$$

Generation & Recombination of carriers :-



$$n \cdot p = n_i^2 = \text{const} \text{ fixed temp}$$

$$n_n \cdot p_n = n_i^2 \rightarrow N \text{ type}$$

$$p_p \cdot n_p = n_i^2 \rightarrow P \text{ type}$$

1/ Generation of carriers :-

- The process by which thermally free e^- s and holes are generated in pairs is called generation of carriers.
- When e^- & h^+ in VB get enough energy, then they will absorb this energy & jump into the conduction band.
- The e^- which is jumped into CB is called thermally generated free e^- and the place from where e^- left is called hole. So, two types of carriers are generated — e^- & holes.

2/ Recombination of carriers :-

- The process of falling back of thermally generated free e^- from conduction band to the hole present in VB is called as recombination of carriers.
- In this process, thermally generated free e^- s and holes are eliminated.

Poisson's Eqⁿ:-

This eqⁿ gives the basic relationship b/w charge and electric field strength. In semiconductor, charges are divided into four components, hole density p , electron density n , acceptor atom density N_A , donor atom density N_D .

$$\frac{d\epsilon}{dx} = \frac{\rho}{\epsilon} = \frac{q}{\epsilon} (p - n - N_D^- + N_A^+)$$

we have diffusion current densities as:-

$$J_n = q D_n n \frac{dn}{dx} + q n \epsilon$$

$$J_p = q D_p p \frac{dp}{dx} - q p \epsilon$$

Continuity Eqⁿ:-

$$\frac{dn}{dt} = \frac{1}{q} \frac{dJ_n}{dx} - (U - G)$$

$$\frac{dp}{dt} = -\frac{1}{q} \frac{dJ_p}{dx} - (U - G)$$

under thermal eqⁿ and

$$\frac{1}{q} \frac{dJ_n}{dx} = U - G$$

$$\frac{1}{q} \frac{dJ_p}{dx} = -(U - G)$$

Diffusion length and mean life time:-

The diffusion length of a carrier type in a material can be defined as the average distance that an excited carrier will travel before recombining.

$$L_D = \sqrt{D\tau}$$

$D \rightarrow$ diffusion coefficient and τ is the lifetime of the excited carrier.

According to Einstein,

$$D = \frac{\mu k_B T}{q}$$

$\mu \rightarrow$ charge carrier mobility

$k_B \rightarrow$ Boltzmann const.

$T \rightarrow$ temperature

$q \rightarrow$ charge of an e⁻.